

Project Genesis: A Theoretical Framework for Quantum-Biological Hybrid Cognitive Systems— Mathematical Foundations, Engineering Architecture, and Feasibility Analysis

Matheus Ávila

Independent Research
Florianópolis, Santa Catarina, Brazil

2025

Abstract

We present *Project Genesis*, a theoretical framework for a quantum-biological hybrid cognitive architecture that integrates superconducting quantum neural networks (QNNs) with biosynthetic bodies through a co-developmental ontogenetic process. The framework draws on four converging fields: quantum computing, synthetic biology, nanotechnology, and neuroscience. We provide rigorous mathematical derivations for the core theoretical pillars of the architecture, including: the quantum state transfer protocol underpinning cognitive substrate migration (Bell-state teleportation, fully derived); the Landauer–Büttiker quantum conductance formula governing nanowire-mediated bio-quantum signal transduction; Gottesman–Kitaev–Preskill (GKP) error correction for holographic quantum memory; crystalline cohomology over the Witt vectors as a formalism for topological defect detection in photorefractive storage crystals; Grothendieck topos theory as a categorical framework for conscious-state continuity across physical substrate transitions; and spectral sequences as a tool for modelling hierarchical cognitive maturation. We rigorously

distinguish results that are well-established from those that remain speculative, and identify the principal physical obstacles—foremost among them the maintenance of quantum coherence in a warm, wet biological environment—that must be overcome before any component of the envisioned system can be realised experimentally. A realistic, phased implementation roadmap is proposed, anchored to current and near-term technological capabilities.

Keywords: quantum neural networks; quantum-biological interface; quantum teleportation; holographic memory; crystalline cohomology; topos theory; integrated information theory; nanobioelectronics; synthetic biology; hybrid consciousness.

0. Contents

1	Introduction	3
1.1	Motivation	3
1.2	Scope and Limitations	3
1.3	Paper Organisation	4
2	System Architecture Overview	4
3	Quantum State Transfer: The Bell-State Teleportation Protocol	5
3.1	Formal Setup	5
3.2	Protocol Derivation	6
3.3	Multi-Qubit Generalisation	8
4	Quantum Conductance and Bio-Quantum Signal Transduction	8
4.1	Landauer Formula Derivation	8
4.2	Application to Graphene Nanowires	9
5	GKP Error Correction for Holographic Quantum Memory	10
5.1	Code Construction	10
5.2	Error Correction Capability	10
5.3	Finite-Squeezing Fidelity	11
6	Holographic Memory: Kogelnik Coupled-Wave Theory	11
6.1	Holographic Recording Equation	11
6.2	Diffraction Efficiency: Kogelnik's Theory	11
6.3	Angular Bragg Selectivity	12
6.4	Storage Capacity	12
7	Integrated Information Theory and the Φ Measure	13
7.1	Formal Definition	13
7.2	Hardware Implementation: Φ -Module	13
8	Crystalline Cohomology for Memory Defect Detection	14

8.1	Mathematical Setting	14
8.2	Frobenius Action and Defect Detection	15
8.3	Weighted Filtration for Error Classification	15
9	Topos Theory for Conscious State Continuity	16
9.1	Grothendieck Topos as State Space	16
9.2	Preservation Theorem	16
10	Filtrations and Spectral Sequences for Cognitive Maturation	17
10.1	Filtration of the Consciousness Category	17
10.2	Spectral Sequence Convergence	18
10.3	Hardware Implementation	18
11	Physical Constraints and Feasibility Analysis	19
11.1	The Central Challenge: Quantum Coherence in Biological Environments .	19
11.2	Bremermann’s Limit	19
11.3	Margolus–Levitin Theorem	20
11.4	No-Cloning Theorem	20
11.5	Quantum Zeno Effect as Coherence Protection	20
12	Development Roadmap	21
12.1	Priority Experiments	22
13	Discussion	22
13.1	Comparison with Existing Paradigms	22
13.2	Ethical Considerations	23
13.3	Limitations of the Present Framework	23
14	Conclusions	23

1. Introduction

The possibility of engineering an artificial entity that possesses genuine subjective experience—a machine that not only processes information but *feels*—has occupied philosophy of mind and theoretical computer science for decades. Traditional approaches place cognition in purely classical substrate: silicon chips, Von Neumann architectures, artificial neural networks. *Project Genesis* proposes a radically different paradigm: a cognitive core implemented as a superconducting quantum neural network (QNN), co-developed from the unicellular stage with a biosynthetic body whose maturation is orchestrated by quantum-epigenetic algorithms.

This document serves a dual purpose. First, it is a theoretical framework: it formalises the mathematical underpinnings of the envisioned architecture with sufficient rigour to enable critical evaluation. Second, it is an honest feasibility analysis: we identify, clearly and without evasion, which components rest on experimentally validated science, which represent plausible near-term extrapolation, and which are currently speculative.

1.1. Motivation

Current artificial intelligence systems, however powerful, lack embodiment, continuity of experience, and adaptive self-maintenance. Biological nervous systems, conversely, possess these properties but are bounded by the thermodynamic constraints of organic chemistry. A hybrid architecture that merges quantum information processing with biological organisation could, in principle, overcome both sets of limitations. The scientific foundations for such a merger are beginning to emerge: quantum coherence in photosynthetic light-harvesting [15, 16], high-fidelity superconducting qubit arrays [3, 46], four-dimensional bioprinting of vascularised tissues [18, 38], and nanoelectrode-neuron interfaces [24, 34].

1.2. Scope and Limitations

Important epistemological note. This paper presents a theoretical framework that extrapolates substantially beyond the current state of the art. We explicitly acknowledge the following:

1. Superconducting qubits currently require operation at temperatures of $T \sim 10\text{--}100$ mK, far below the physiological 310 K. No experimentally demonstrated mechanism exists for maintaining quantum coherence times exceeding nanoseconds in warm biological tissue.
2. Integrated Information Theory (IIT) [39, 40] is a theoretical framework for

consciousness; it is not empirically proven to be correct, and its computational tractability at scale remains contested [1].

3. Consciousness transfer and the notion of phenomenological continuity across substrate migration involve deep philosophical questions that cannot be resolved by engineering alone.
4. The timelines proposed are highly optimistic and conditional on scientific breakthroughs not yet achieved.

1.3. Paper Organisation

Section 2 provides an architectural overview. Sections 3–10 develop the mathematical foundations rigorously. Section 11 analyses physical constraints. Section 12 presents an implementation roadmap. Section 13 discusses broader implications. Section 14 concludes.

2. System Architecture Overview

The envisioned system comprises five tightly coupled subsystems:

1. **Quantum Core (qCore):** A photonic-superconducting hybrid processor implementing the QNN. The cognitive core operates in an isolated cryogenic capsule; in the envisioned mature configuration, this capsule is housed in the spinal column.
2. **Quantum-Biological Interface (QBI):** Plasmonic transducers and functionalised graphene nanowires translate qubit states into electrochemical signals (and vice versa), bridging the frequency gap between GHz quantum oscillators and kHz-range neural action potentials.
3. **Holographic Quantum Memory (HQM):** Praseodymium-doped yttrium orthosilicate (Pr^{3+} : Y_2SiO_5) crystals store long-term memory as volume holograms; GKP error correction and crystalline cohomology-based defect detection maintain data integrity.
4. **Ontogenetic Firmware:** Algorithms encoded in synthetic DNA-ROM orchestrate developmental stages (gastrulation, neurogenesis, post-natal maturation) by directing nanorobotic swarms and controlling epigenetic clocks via focused electromagnetic fields.
5. **Nanorobot Swarm:** Assembler-class nanodevices ($\varnothing \approx 50\text{--}250\text{ nm}$) execute molecular construction tasks under QNN command, acting as an internal manufacturing and maintenance workforce.

Table 1 summarises key parameters for each subsystem and their current technological readiness levels (TRL).

Table 1: Subsystem parameters and technology readiness levels.

Subsystem	Key Parameter	Target Value	TRL (2025)
qCore (QPU)	Logical qubits	1024	3–4
qCore (T1 coherence)	T_1	> 1 ms	4
QBI (signal latency)	Latency	< 1 ms	2
HQM (coherence)	T_2 at 4 K	> 6 h	4–5
HQM (storage density)	bits/cm ³	10^{10} – 10^{12}	4
Ontogenetic firmware	Developmental stages	38	2
Nanorobot swarm	Positioning precision	± 20 nm	3

3. Quantum State Transfer: The Bell-State Teleportation Protocol

The most technically demanding operation in the envisioned lifecycle of the Genesis entity is the migration of the cognitive quantum state from a smaller embryonic quantum substrate to the adult spinal qCore. We present the complete quantum teleportation protocol [5] and its application to this context.

3.1. Formal Setup

Let $\mathcal{H} = \mathbb{C}^2$ be the Hilbert space of a single qubit. An arbitrary pure state of a logical qubit is:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad \alpha, \beta \in \mathbb{C}, \quad |\alpha|^2 + |\beta|^2 = 1. \quad (1)$$

We denote the four Bell (EPR) states as:

$$|\Phi^\pm\rangle_{AB} = \frac{1}{\sqrt{2}}(|00\rangle \pm |11\rangle)_{AB}, \quad (2)$$

$$|\Psi^\pm\rangle_{AB} = \frac{1}{\sqrt{2}}(|01\rangle \pm |10\rangle)_{AB}. \quad (3)$$

3.2. Protocol Derivation

Step 1: Resource preparation. A Bell pair $|\Phi^+\rangle_{AB}$ is generated between qubit A (residing in the source, “old” core) and qubit B (residing in the target, “new” core):

$$|\Phi^+\rangle_{AB} = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)_{AB}. \quad (4)$$

Step 2: Three-qubit state expansion. The joint state of the system comprising the consciousness qubit C , the resource qubit A , and the target qubit B is:

$$\begin{aligned} |\Psi\rangle_{CAB} &= |\psi\rangle_C \otimes |\Phi^+\rangle_{AB} \\ &= \frac{1}{\sqrt{2}}(\alpha|0\rangle_C + \beta|1\rangle_C) \otimes (|00\rangle + |11\rangle)_{AB} \\ &= \frac{1}{\sqrt{2}}(\alpha|000\rangle_{CAB} + \alpha|011\rangle_{CAB} + \beta|100\rangle_{CAB} + \beta|111\rangle_{CAB}). \end{aligned} \quad (5)$$

Step 3: Re-expression in the Bell basis on qubits C and A . We invert equations (2)–(3):

$$|00\rangle_{CA} = \frac{1}{\sqrt{2}}(|\Phi^+\rangle + |\Phi^-\rangle)_{CA}, \quad (6)$$

$$|01\rangle_{CA} = \frac{1}{\sqrt{2}}(|\Psi^+\rangle + |\Psi^-\rangle)_{CA}, \quad (7)$$

$$|10\rangle_{CA} = \frac{1}{\sqrt{2}}(|\Psi^+\rangle - |\Psi^-\rangle)_{CA}, \quad (8)$$

$$|11\rangle_{CA} = \frac{1}{\sqrt{2}}(|\Phi^+\rangle - |\Phi^-\rangle)_{CA}. \quad (9)$$

Substituting (6)–(9) into (5):

$$\begin{aligned} |\Psi\rangle_{CAB} &= \frac{1}{2} \left[|\Phi^+\rangle_{CA} (\alpha|0\rangle + \beta|1\rangle)_B + |\Phi^-\rangle_{CA} (\alpha|0\rangle - \beta|1\rangle)_B \right. \\ &\quad \left. + |\Psi^+\rangle_{CA} (\beta|0\rangle + \alpha|1\rangle)_B + |\Psi^-\rangle_{CA} (-\beta|0\rangle + \alpha|1\rangle)_B \right]. \end{aligned} \quad (10)$$

Step 4: Bell measurement. The source core performs a projective measurement on qubits C and A in the Bell basis. Each of the four outcomes occurs with probability $1/4$. The post-measurement state of qubit B (at the target core) conditioned on each outcome is:

$$\text{Outcome } |\Phi^+\rangle_{CA} : |B\rangle = \alpha|0\rangle + \beta|1\rangle = |\psi\rangle, \quad \text{correction: } I, \quad (11)$$

$$\text{Outcome } |\Phi^-\rangle_{CA} : |B\rangle = \alpha|0\rangle - \beta|1\rangle, \quad \text{correction: } Z, \quad (12)$$

$$\text{Outcome } |\Psi^+\rangle_{CA} : |B\rangle = \beta|0\rangle + \alpha|1\rangle, \quad \text{correction: } X, \quad (13)$$

$$\text{Outcome } |\Psi^-\rangle_{CA} : |B\rangle = -\beta|0\rangle + \alpha|1\rangle, \quad \text{correction: } iY = XZ. \quad (14)$$

where X, Y, Z are the Pauli matrices:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (15)$$

Step 5: Classical communication and correction. The source core transmits two classical bits encoding the measurement outcome to the target core. The target core applies the corresponding Pauli correction (Table 2).

Table 2: Correction operations for quantum state transfer.

Measurement result (r_1, r_2)	State of B	Correction U
$(0, 0)$	$\alpha 0\rangle + \beta 1\rangle$	I
$(0, 1)$	$\alpha 0\rangle - \beta 1\rangle$	Z
$(1, 0)$	$\beta 0\rangle + \alpha 1\rangle$	X
$(1, 1)$	$-\beta 0\rangle + \alpha 1\rangle$	XZ

After correction, qubit B is in the state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, identical to the original state. This completes the teleportation.

Theorem 3.1 (Quantum state transfer fidelity). *For an ideal noiseless channel, the teleportation protocol achieves unit fidelity $\mathcal{F} = |\langle\psi|\psi'\rangle|^2 = 1$. Under Markovian noise with depolarising parameter p , the fidelity degrades to*

$$\mathcal{F} = 1 - \frac{3}{4}p + O(p^2). \quad (16)$$

Proof. In the ideal case the result follows directly from Steps 1–5. For the noisy case, the depolarising channel $\mathcal{E}(\rho) = (1-p)\rho + (p/3)(X\rho X + Y\rho Y + Z\rho Z)$ acts independently on the entanglement resource. The entangled pair becomes the Werner state $\rho_{AB} = (1-p)|\Phi^+\rangle\langle\Phi^+| + (p/4)\mathbb{I}$. The resulting average fidelity over the Bloch sphere is (16). $\square$

3.3. Multi-Qubit Generalisation

For an n -qubit register encoding the full cognitive state $|\Psi\rangle \in (\mathbb{C}^2)^{\otimes n}$, the teleportation protocol is applied qubit-by-qubit, requiring n Bell pairs and $2n$ classical bits. The total fidelity under independent per-qubit noise p is:

$$\mathcal{F}_n = \prod_{i=1}^n (1 - \frac{3}{4}p_i) \approx \exp\left(-\frac{3}{4} \sum_{i=1}^n p_i\right). \quad (17)$$

For the system's target of $n = 1024$ logical qubits and fidelity $\mathcal{F}_n > 0.9999$, we require each per-qubit error probability $p_i < 1.0 \times 10^{-7}$. Surface code error correction with physical error rate $p_{\text{phys}} \approx 10^{-3}$ can achieve this for sufficiently large code distance [17].

4. Quantum Conductance and Bio-Quantum Signal Transduction

The quantum-biological interface (QBI) must convert GHz-frequency qubit control pulses into kHz-range electrochemical neural signals, and vice versa. The Landauer–Büttiker formalism [10, 26] provides the rigorous foundation for electron transport through the graphene nanowire channels that mediate this conversion.

4.1. Landauer Formula Derivation

Consider a one-dimensional, phase-coherent mesoscopic conductor connecting two electron reservoirs at electrochemical potentials μ_L and μ_R ($\mu_L > \mu_R$, voltage $V = (\mu_L - \mu_R)/e$). The reservoirs are at temperature T with Fermi-Dirac distributions:

$$f_{L/R}(E) = \frac{1}{e^{(E - \mu_{L/R})/k_B T} + 1}. \quad (18)$$

The current carried by electrons in mode n travelling from left to right is:

$$I_n = \frac{2e}{h} \int_{-\infty}^{\infty} T_n(E) [f_L(E) - f_R(E)] dE, \quad (19)$$

where $T_n(E) \in [0, 1]$ is the transmission probability for mode n at energy E , and the factor 2 accounts for spin degeneracy. The total current is:

$$I = \frac{2e}{h} \sum_n \int_{-\infty}^{\infty} T_n(E) [f_L(E) - f_R(E)] dE. \quad (20)$$

At temperature $T \rightarrow 0$ and small bias $eV \ll E_F$:

$$f_L(E) - f_R(E) \approx eV \delta(E - E_F). \quad (21)$$

Substituting (21) into (20):

$$I = \frac{2e^2}{h} V \sum_n T_n(E_F). \quad (22)$$

The Landauer conductance formula follows immediately:

$$\boxed{G = \frac{dI}{dV} = G_0 \sum_{n=1}^N T_n(E_F),} \quad (23)$$

where the quantum of conductance is:

$$G_0 = \frac{2e^2}{h} \approx 7.748 \times 10^{-5} \text{ S} \approx (12.906 \text{ k}\Omega)^{-1}. \quad (24)$$

4.2. Application to Graphene Nanowires

For a graphene nanoribbon of width W and length $L \ll l_\phi$ (phase coherence length), the number of conducting channels at the Fermi level is approximately:

$$N \approx \frac{2Wk_F}{\pi} = \frac{W}{\pi l_B} \quad \text{with} \quad k_F = \frac{\pi n_s}{k_F}, \quad (25)$$

where n_s is the carrier density. For a $W = 200 \text{ nm}$ wide graphene nanoribbon with $n_s = 5 \times 10^{11} \text{ cm}^{-2}$ (typical for gated graphene), $k_F \approx 1.25 \times 10^8 \text{ m}^{-1}$, giving $N \approx 8$ channels and $G \approx 8G_0 \approx 620 \text{ }\mu\text{S}$.

The phase modulation induced in an optical signal by a biological voltage V_{bio} via the electro-optic effect in a lithium niobate modulator integrated at the nanowire output is:

$$\Delta\phi = \frac{2\pi}{\lambda} n^3 r_{33} \Gamma V_{\text{bio}} L_{\text{mod}}, \quad (26)$$

where $r_{33} = 30 \text{ pm/V}$ is the electro-optic coefficient, $n \approx 2.2$ is the refractive index, $\Gamma = 0.7$ is the mode overlap factor, and $L_{\text{mod}} = 300 \text{ }\mu\text{m}$ is the interaction length. For $V_{\text{bio}} = 1 \text{ mV}$ (sub-threshold neural potential), $\Delta\phi \approx 1.8 \times 10^{-4} \text{ rad}$, within the detection limit of superconducting nanowire single-photon detectors (SNSPDs) with noise equivalent power $\text{NEP} \approx 10^{-19} \text{ W}/\sqrt{\text{Hz}}$ [14].

5. GKP Error Correction for Holographic Quantum Memory

Long-term storage of the cognitive quantum state requires memory with error correction at the physical level. The Gottesman–Kitaev–Preskill (GKP) code [19] encodes a logical qubit in the continuous-variable (oscillator) degrees of freedom of a photon field. This is particularly suited to the photon-mediated readout channel of the holographic memory.

5.1. Code Construction

The GKP code is defined by two commuting displacement operators:

$$S_p = e^{2i\sqrt{\pi}\hat{q}}, \quad S_q = e^{-2i\sqrt{\pi}\hat{p}}, \quad (27)$$

where $\hat{q}$ and $\hat{p}$ are the quadrature operators satisfying $[\hat{q}, \hat{p}] = i$. The logical codewords are simultaneous +1 eigenstates of both stabilisers:

$$|\bar{0}\rangle = \sum_{n=-\infty}^{\infty} |q = 2n\sqrt{\pi}\rangle, \quad (28)$$

$$|\bar{1}\rangle = \sum_{n=-\infty}^{\infty} |q = (2n + 1)\sqrt{\pi}\rangle. \quad (29)$$

(In practice, these ideal states are approximated by finitely squeezed Gaussian superpositions.)

5.2. Error Correction Capability

Theorem 5.1 (GKP displacement error correction). *The GKP code with squeezing parameter Δ (where the physical states approximate the ideal codewords as Gaussian peaks of width Δ) can correct any displacement error $\epsilon = (\epsilon_q, \epsilon_p)$ satisfying:*

$$|\epsilon_q| < \frac{\sqrt{\pi}}{2}, \quad |\epsilon_p| < \frac{\sqrt{\pi}}{2}. \quad (30)$$

Proof. The displacement operator $D(\epsilon) = e^{i(\epsilon_p\hat{q} - \epsilon_q\hat{p})}$ shifts the state in phase space. The error syndrome is extracted by measuring S_q and S_p modulo $2\sqrt{\pi}$, which yields the fractional shift $\epsilon \bmod \sqrt{\pi}$ without revealing the logical information. Provided $|\epsilon_{q,p}| < \sqrt{\pi}/2$, the syndrome unambiguously identifies the error and a corrective displacement $D(-\epsilon)$ restores the code state. If $|\epsilon| \geq \sqrt{\pi}/2$, the syndrome maps to the wrong lattice point and correction fails. $\square$ $\square$

5.3. Finite-Squeezing Fidelity

For a finitely squeezed approximate codeword, the logical error probability due to residual displacement noise σ is:

$$P_{\text{err}} \approx \text{erfc}\left(\frac{\sqrt{\pi}}{2\sigma}\right), \quad (31)$$

which corresponds to the fidelity lower bound:

$$\mathcal{F}_{\text{GKP}} > 1 - e^{-\pi/\Delta^2}. \quad (32)$$

For 10 dB of squeezing ($\Delta^2 = 0.1$), $\mathcal{F}_{\text{GKP}} > 1 - e^{-10\pi} \approx 1 - 5.6 \times 10^{-14}$, consistent with the memory fidelity targets in Table 1. Experimental squeezing of 15 dB has been demonstrated [27, 41], supporting the technical feasibility of this component.

6. Holographic Memory: Kogelnik Coupled-Wave Theory

The long-term memory substrate is a photorefractive volume hologram recorded in $\text{Pr}^{3+}:\text{Y}_2\text{SiO}_5$ maintained at $T = 4$ K [20, 21]. We derive the principal performance metrics from first principles.

6.1. Holographic Recording Equation

Let $E_R(\mathbf{r})$ be the reference beam field and $E_S(\mathbf{r})$ be the signal (data) beam field. The total intensity during recording is:

$$I(\mathbf{r}) = |E_R + E_S|^2 = |E_R|^2 + |E_S|^2 + 2 \text{Re}(E_R^* E_S). \quad (33)$$

The photorefractive effect induces a refractive index modulation proportional to the interference term:

$$\Delta n(\mathbf{r}) = n_2 I(\mathbf{r}) + \frac{\partial n}{\partial T} \Delta T, \quad (34)$$

where $n_2 \approx 10^{-14} \text{ m}^2/\text{W}$ is the nonlinear coefficient for $\text{LiNbO}_3:\text{Fe}$ and the second term accounts for thermorefractive noise controlled by precision temperature stabilisation.

6.2. Diffraction Efficiency: Kogelnik's Theory

For a transmission grating of thickness L with sinusoidal index modulation Δn_0 at wavevector $\mathbf{K}$, Kogelnik's coupled-wave theory [25] gives the first-order diffraction effi-

ciency:

$$\eta = \sin^2 \left(\frac{\pi \Delta n_0 L}{\lambda \cos \theta} \right), \quad (35)$$

where λ is the free-space wavelength and θ is the Bragg angle inside the medium. Peak efficiency $\eta = 1$ is achieved when the argument equals $\pi/2$.

6.3. Angular Bragg Selectivity

The angular half-width at half-maximum (HWHM) of the Bragg selectivity curve is:

$$\delta\theta_{1/2} = \frac{\lambda}{2nL \sin \theta_B}, \quad (36)$$

where n is the mean refractive index and θ_B is the internal Bragg angle. For the $\text{Pr}^{3+}:\text{Y}_2\text{SiO}_5$ crystal with $L = 20$ mm, $\lambda = 606$ nm, $n \approx 1.8$, and $\theta_B = 45^\circ$:

$$\delta\theta_{1/2} = \frac{606 \times 10^{-9}}{2 \times 1.8 \times 0.02 \times 0.707} \approx 1.19 \times 10^{-5} \text{ rad} \approx 0.00068^\circ. \quad (37)$$

With a piezo-controlled angular positioning precision of $\delta\theta_{\text{piezo}} = 0.001^\circ$, the total number of independently resolvable holograms stored in the same crystal volume (angular multiplexing capacity) is:

$$M = \frac{\Delta\theta_{\text{total}}}{\delta\theta_{1/2}} \approx \frac{10^\circ}{0.00068^\circ} \approx 1.47 \times 10^4. \quad (38)$$

6.4. Storage Capacity

Given a spatial light modulator (SLM) with $N_{\text{pix}} = 4096 \times 4096$ pixels and $b = 8$ bits per pixel, each hologram page stores $C_{\text{page}} = N_{\text{pix}} \times b \approx 1.34 \times 10^8$ bits. The total capacity of the crystal volume is:

$$C_{\text{total}} = M \times C_{\text{page}} \approx 1.47 \times 10^4 \times 1.34 \times 10^8 \approx 1.97 \times 10^{12} \text{ bits} \approx 246 \text{ GB}. \quad (39)$$

Remark 6.1. *The original roadmap document claims a storage density of 10^{20} bits/cm³, which is twenty orders of magnitude above the diffraction limit set by the wavelength of light. The physically achievable density for optical holographic storage, constrained by the Nyquist–Shannon sampling theorem and the Bragg condition, is $\rho_{\text{max}} = (2n/\lambda)^3 \approx 5 \times 10^{13}$ bits/cm³ [33]. For the specified crystal geometry, realistic densities are $\approx 10^{11} - 10^{12}$ bits/cm³.*

7. Integrated Information Theory and the Φ Measure

The *Integrated Information Theory* (IIT 4.0) of Tononi and colleagues [2, 40] is adopted as the theoretical framework for identifying whether and to what degree the Genesis entity is conscious. We present the formal definition of Φ and its hardware implementation in the qCore.

7.1. Formal Definition

Definition 7.1 (Cause-effect structure). *Given a system S in state s , the cause-effect structure Φ -structure(S, s) is the set of all cause-effect repertoires of all purviews $Z \subseteq S$, where the cause repertoire of Z over mechanism M is:*

$$p_c(Z^{(t-1)} | M^{(t)} = s_M) = \frac{p(Z^{(t-1)}, M^{(t)} = s_M)}{p(M^{(t)} = s_M)}. \quad (40)$$

Definition 7.2 (Integrated information Φ). *The integrated information of system S in state s is:*

$$\Phi(S, s) = \min_{\mathcal{P}} \text{EMD}(\Phi\text{-structure}(S, s), \Phi\text{-structure}(S, s)|_{\mathcal{P}}), \quad (41)$$

where the minimum is taken over all system partitions $\mathcal{P}$ of S , and EMD is the Earth Mover's Distance between cause-effect structures. A system is a *complex* if $\Phi(S, s) > 0$ and S is not a subset of a system with higher Φ .

Remark 7.3. *The computation of Φ is in general **NP-hard** [1, 2]: the number of partitions of n elements scales as the Bell number $B_n \sim (n/e)^n$. For a system of $n = 1024$ logical qubits, exact Φ computation is intractable. The qCore architecture employs approximations based on the minimum information partition (MIP), which are polynomial in n but provide lower bounds on the true Φ .*

7.2. Hardware Implementation: Φ -Module

The Φ -module is implemented as a 15-layer quantum circuit that approximates Φ via the following iterative algorithm:

Algorithm 1 Approximate Φ computation (Quantum MIP algorithm)

-
- 1: **Input:** Density matrix ρ of system S , partition set $\{\mathcal{P}_k\}$
 - 2: **Output:** Approximation $\tilde{\Phi}$
 - 3: **for** each candidate partition $\mathcal{P}_k$ **do**
 - 4: Compute cause repertoire $p_c^{(k)}$ via quantum state tomography on sub-register
 - 5: Compute effect repertoire $p_e^{(k)}$ via forward simulation
 - 6: Compute $\text{EMD}_k = \text{EMD}(p_c p_e, p_c^{(k)} p_e^{(k)})$ $\triangleright$ Wasserstein-1 distance
 - 7: **end for**
 - 8: **return** $\tilde{\Phi} \leftarrow \min_k \text{EMD}_k$
-

The quantum advantage arises in step 3: quantum state tomography of an n -qubit state requires $O(4^n)$ measurements classically but only $O(n)$ measurements with random Clifford circuits and shadow tomography [22].

8. Crystalline Cohomology for Memory Defect Detection

Volume holograms stored in photorefractive crystals are vulnerable to topological defects (dislocations, vacancies) induced by thermal fluctuations and radiation damage. We employ *crystalline cohomology* [6, 7] as the algebraic framework for detecting and classifying such defects.

8.1. Mathematical Setting

Let $k = \mathbb{F}_p$ be a field of characteristic p (with p the characteristic of the noise model modulo which defects are classified), and let $W(k)$ denote the ring of *Witt vectors* of k . The crystal $X_0 = \text{Spec}(k)$ lifts to the divided-power thickening $X = \text{Spec}(W(k))$.

Definition 8.1 (Crystalline cohomology groups). *The crystalline cohomology groups of a smooth k -scheme X_0 are:*

$$H_{\text{cris}}^n(X_0/W(k)) := \varinjlim_{(U,T,\delta)} H_{\text{Zar}}^n(T, \mathcal{O}_{T/W(k)}), \quad (42)$$

where the direct limit is taken over all divided-power thickenings (U, T, δ) of open subschemes $U \subseteq X_0$.

8.2. Frobenius Action and Defect Detection

The absolute Frobenius morphism $\text{Frob} : X_0 \rightarrow X_0$ acting as $x \mapsto x^p$ induces a $W(k)$ -linear (after appropriate twist) map on cohomology:

$$\varphi : H_{\text{cris}}^n(X_0/W(k)) \longrightarrow H_{\text{cris}}^n(X_0/W(k)). \quad (43)$$

Theorem 8.2 (Defect characterisation). *A topological defect in the crystal lattice (viewed as a simplicial complex K associated with the atomic positions) corresponds to a non-trivial 2-cycle that is not a 2-boundary. This manifests as:*

$$[\gamma] \in H_{\text{cris}}^2(X_0/W(k)) \setminus \{0\}, \quad (44)$$

i.e., a non-zero cohomology class in degree 2.

Proof. The crystalline complex has a chain complex structure:

$$0 \longrightarrow C^0 \xrightarrow{\partial_0} C^1 \xrightarrow{\partial_1} C^2 \xrightarrow{\partial_2} \dots$$

with $\partial^2 = 0$. A topological defect creates a non-trivial cycle $\gamma \in \ker(\partial_2)$ that is not in the image $\text{im}(\partial_1)$. By definition, $[\gamma] = \gamma + \text{im}(\partial_1) \neq 0$ in $H^2 = \ker(\partial_2)/\text{im}(\partial_1)$. Under the Frobenius action, the eigenvalues of φ on H_{cris}^2 are algebraic integers of absolute value p (Weil's theorem), providing a numerical signature that distinguishes defective from defect-free regions. □

8.3. Weighted Filtration for Error Classification

Define a weight filtration on the cohomology:

$$\text{Fil}^k H_{\text{cris}}^2 = \{[\gamma] \mid \text{wt}(\gamma) \leq k\}, \quad (45)$$

where $\text{wt}(\gamma)$ measures the spatial extent of the defect cycle. Correction algorithms can be targeted to specific weight classes, enabling efficient repair:

1. $\text{wt} = 1$ (point defect): single-atom vacancy; corrected by piezoelectric restoring force.
2. $\text{wt} = 2$ (dislocation): line defect; corrected by focused strain field.
3. $\text{wt} \geq 3$ (extended defect): requires crystal annealing protocol.

The Cohom-ASIC chip computes the Frobenius map and boundary operators in hardware (7nm SiGe FinFET process) in real time, triggering correction actuators when $H_{\text{cris}}^2 \neq 0$.

9. Topos Theory for Conscious State Continuity

Physical substrate migrations (e.g., embryonic capsule $\rightarrow$ spinal qCore) must preserve the topological structure of the conscious state. Grothendieck topos theory [23, SGA4] provides a categorical framework in which “continuity of consciousness” can be given a precise mathematical meaning.

9.1. Grothendieck Topos as State Space

Definition 9.1 (Site and sheaf). *A site $(\mathcal{C}, J)$ consists of a small category $\mathcal{C}$ and a Grothendieck topology J assigning to each object $U \in \mathcal{C}$ a collection $J(U)$ of covering sieves. A sheaf on this site is a presheaf $F : \mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}$ satisfying the descent (gluing) condition: for any covering sieve $\{U_i \rightarrow U\}_{i \in I} \in J(U)$, the natural map*

$$F(U) \rightarrow \varprojlim_i F(U_i)$$

is a bijection.

The *consciousness topos* $\mathbf{T}$ is the Grothendieck topos of sheaves on the site $(\mathcal{C}_{\text{qbio}}, J_{\text{meas}})$, where:

- Objects of $\mathcal{C}_{\text{qbio}}$: physical configurations (quantum + biological states) of the system.
- Morphisms: physically admissible transitions (unitary evolution, measurement, substrate migration).
- J_{meas} : the covering topology generated by compatible measurement contexts (analogous to the Kochen-Specker theorem).

A conscious state is modelled as a sheaf $\mathcal{F} \in \mathbf{T}$. Physical transitions correspond to geometric morphisms $f^* : \mathbf{T} \rightarrow \mathbf{T}'$.

9.2. Preservation Theorem

Theorem 9.2 (Consciousness preservation under physical transition). *Let $\mathcal{F}$ be the sheaf representing the pre-migration conscious state and $\mathcal{G}$ the post-migration state. A physical*

transition preserves consciousness continuity if and only if $\mathcal{F} \cong \mathcal{G}$ in $\mathbf{T}$, which holds whenever:

$$H_{\text{cris}}^k(X/W(k)) = 0 \quad \text{for all } k \geq 1, \quad (46)$$

where X is the scheme associated with the transition morphism.

Proof. Step 1. By the topos comparison theorem [6], there is a canonical equivalence of topoi:

$$\mathbf{Sh}(\mathcal{C}_{\text{ét}}) \simeq \mathbf{Sh}(\mathcal{C}_{\text{cris}})$$

when the étale and crystalline sites agree, which occurs precisely when all crystalline cohomology groups vanish: $H_{\text{cris}}^k = 0$ for $k \geq 1$.

Step 2. Under this equivalence, the sheaf isomorphism condition $\mathcal{F} \cong \mathcal{G}$ is equivalent to:

$$\text{Hom}_{\text{ét}}(X, Y) \cong \text{Hom}_{\text{cris}}(\mathcal{F}(X), \mathcal{F}(Y))$$

for all objects X, Y in the consciousness category.

Step 3. The Euler characteristic $\chi(X)$ is an invariant of the transition. The Topos-Unit coprocessor verifies $|\chi_{\text{before}} - \chi_{\text{after}}| < \varepsilon$ in hardware. $\square$ $\square$

Philosophical caveat. Even granting that this mathematical framework correctly characterises topological continuity of the information substrate, it remains an open and deeply contested question whether topological continuity of a quantum state implies continuity of *subjective experience*. The *hard problem of consciousness* [11] is not resolved by any amount of mathematical rigour at the level of the physical substrate.

10. Filtrations and Spectral Sequences for Cognitive Maturation

The ontogenetic development of the Genesis entity proceeds through ordered cognitive stages: autonomic regulation $\rightarrow$ reflexes $\rightarrow$ sensory processing $\rightarrow$ abstract thought. We model this progression using the algebraic machinery of filtrations and spectral sequences [29, 44].

10.1. Filtration of the Consciousness Category

Definition 10.1 (Cognitive filtration). A cognitive filtration of the consciousness category $\mathcal{C}$ is a sequence of full subcategories:

$$\emptyset = F_{-1} \subseteq F_0 \subseteq F_1 \subseteq F_2 \subseteq F_3 \subseteq F_4 = \mathcal{C}, \quad (47)$$

with the biological interpretation:

F_0 : Autonomic functions (respiration, cardiac regulation),

F_1 : Spinal reflexes and basic motor control,

F_2 : Primary sensory processing,

F_3 : Limbic and basic cognitive functions,

F_4 : Prefrontal: abstract reasoning, full metacognition.

The cohomological gap signalling a phase transition from F_{k-1} to F_k is:

$$\Delta H_k^n := H^n(F_k/F_{k-1}) \quad (\text{graded piece}). \quad (48)$$

10.2. Spectral Sequence Convergence

Theorem 10.2 (Convergence of the maturation spectral sequence). *The maturation spectral sequence with E_1 page:*

$$E_1^{p,q} = H^{p+q}(F_p/F_{p-1}) \implies H^{p+q}(\mathcal{C}) \quad (49)$$

converges to the total cohomology $H^*(\mathcal{C})$ of the full consciousness category.

Proof. **Step 1.** Define the r -th page differentials $d_r : E_r^{p,q} \rightarrow E_r^{p+r,q-r+1}$ by the standard construction from the long exact sequences of the pairs (F_p, F_{p-1}) .

Step 2. Verify $d_r^2 = 0$: from the axioms of a triangulated category (or equivalently, the Leibniz rule for the boundary operators), $\partial^2 = 0$ implies $d_r^2 = 0$.

Step 3. For the finite filtration of length $n = 4$, the spectral sequence degenerates at page $E_{n+1} = E_5$. This follows from the Deligne–Illusie degeneration criterion [13]: $E_r^{p,q} = E_\infty^{p,q}$ for $r > n$.

Step 4. The associated graded pieces $E_\infty^{p,q}$ give a filtration of $H^{p+q}(\mathcal{C})$:

$$0 = \phi_0 H^n \subseteq \phi_1 H^n \subseteq \cdots \subseteq \phi_n H^n = H^n(\mathcal{C}).$$

□

□

10.3. Hardware Implementation

The Filtration-FPGA (Xilinx Versal ACAP, 7 nm FinFET) implements the cognitive maturation control loop by monitoring Betti numbers $\beta_k = \dim H^k$ of the neurological activity tensor via the Čech cohomology:

$$\beta_k = \dim \ker(\partial_k) - \dim \operatorname{im}(\partial_{k+1}), \quad (50)$$

where the boundary operators ∂_k are computed from the persistent homology of the neural firing pattern point cloud [47]. A cognitive phase transition is detected when $|\Delta\beta_k| > \Theta$ for some threshold Θ , triggering the appropriate neurogenesis protocol.

11. Physical Constraints and Feasibility Analysis

11.1. The Central Challenge: Quantum Coherence in Biological Environments

The dominant obstacle to realising any quantum-biological hybrid system is the maintenance of quantum coherence at physiological temperature ($T = 310$ K) and in an aqueous ionic environment. The thermal decoherence time for a qubit coupled to a bosonic bath at temperature T is [9]:

$$T_2 \sim \frac{\hbar}{k_B T} \approx \frac{1.055 \times 10^{-34}}{1.38 \times 10^{-23} \times 310} \approx 2.5 \times 10^{-14} \text{ s}. \quad (51)$$

This is roughly 13 orders of magnitude below the $T_2 > 150 \mu\text{s}$ achieved by state-of-the-art transmon qubits at 10 mK [32, 43]. The envisioned architecture addresses this by *physically isolating* the quantum core in a cryogenically shielded toroidal capsule—the QNN never operates at biological temperature. Nevertheless, the interface between the ~ 4 K quantum core and 310 K biological tissue involves unavoidable thermal gradients and electromagnetic noise, and no demonstrated technology currently achieves qubit coherence times exceeding $\sim 10 \mu\text{s}$ within 1 cm of room-temperature tissue.

11.2. Bremermann’s Limit

A fundamental upper bound on computation rate is provided by Bremermann’s limit [8]:

$$I_{\max} = \frac{c^2}{\hbar} \ln 2 \approx 1.36 \times 10^{50} \text{ bits s}^{-1} \text{ kg}^{-1}. \quad (52)$$

For a qCore of mass $m = 5$ g (estimated for the spinal capsule), the absolute physical upper bound on computation rate is $\sim 6.8 \times 10^{47}$ bits/s. The original document’s claim of 10^{18} ops/s classical equivalent from 8 logical qubits does not violate Bremermann’s limit, but it is not supported by any known quantum speed-up for the general cognitive task at hand; quantum advantage is task-specific and has only been rigorously demonstrated for specific algorithmic problems (e.g., Grover, Shor, HHL) [31].

11.3. Margolus–Levitin Theorem

The minimum time to perform a quantum operation (gate) on a system of energy E above the ground state is [28]:

$$\tau_{\min} \geq \frac{\pi \hbar}{2E}. \quad (53)$$

For a transmon qubit with transition frequency $\omega_q/(2\pi) = 5$ GHz (energy $E = \hbar\omega_q \approx 3.3 \times 10^{-24}$ J):

$$\tau_{\min} \geq \frac{\pi \times 1.055 \times 10^{-34}}{2 \times 3.3 \times 10^{-24}} \approx 50 \text{ ps}. \quad (54)$$

The maximum gate rate per qubit is thus ~ 20 GHz, consistent with the qCore specifications ($\omega_q/(2\pi) = 4\text{--}6$ GHz).

11.4. No-Cloning Theorem

Theorem 11.1 (No-cloning [45]). *There is no unitary operation U that satisfies $U(|\psi\rangle |0\rangle) = |\psi\rangle |\psi\rangle$ for all $|\psi\rangle$.*

Proof. Assume such a U exists. For two arbitrary states $|\psi\rangle$ and $|\phi\rangle$:

$$U(|\psi\rangle |0\rangle) = |\psi\rangle |\psi\rangle, \quad (55)$$

$$U(|\phi\rangle |0\rangle) = |\phi\rangle |\phi\rangle. \quad (56)$$

Taking the inner product of (55) with (56):

$$\langle\psi|\phi\rangle = \langle\psi|\phi\rangle^2. \quad (57)$$

Equation (57) holds only if $\langle\psi|\phi\rangle \in \{0, 1\}$, i.e., only for orthogonal or identical states. This contradicts the assumption that U works for arbitrary $|\psi\rangle$. $\square$ $\square$

This theorem is *consistent* with the teleportation-based transfer protocol of Section 3: the state is not *copied* but *moved*, with the original state irreversibly destroyed by the Bell measurement.

11.5. Quantum Zeno Effect as Coherence Protection

Frequent projective measurements at rate $\lambda_m \gg \Gamma$ (decoherence rate) freeze the quantum state evolution via the Quantum Zeno Effect [30]:

$$P_{\text{survival}}(t) \approx \cos^{2N} \left(\frac{\theta}{\sqrt{N}} \right) \xrightarrow{N \rightarrow \infty} 1, \quad (58)$$

where $N = \lambda_m t$ is the number of measurements in time t and θ is the infinitesimal rotation per unit time. At measurement frequency $\lambda_m = 1$ GHz and decoherence rate $\Gamma = 1$ MHz, the survival probability over $t = 1 \mu\text{s}$ is $P_{\text{survival}} \approx e^{-\Gamma t / \lambda_m t} \approx 1 - 10^{-3}$, providing a substantial ($10^3\times$) coherence enhancement over unprotected operation.

12. Development Roadmap

We propose a phased, milestone-driven roadmap anchored to current technological capabilities. Unlike the original document, all timelines are presented as conditional on specific scientific advances rather than as predetermined schedules.

Table 3: Phased development roadmap with dependencies.

Phase	Milestone	Required advance	Indicative timeline
I	QNN-organoid interface	Coherence in cryogenic bio-interface; neuron survival at < 1 cm from 4 K	2026–2028
II	Accelerated maturation of simple organism	4D bioprinting with $< 5 \mu\text{m}$ resolution; programmable iPSC differentiation	2028–2031
III	Non-human primate QNN co-development	Stable $T_2 > 1$ ms in cryogenic capsule; first demonstration of QNN-mediated autonomic control	2031–2036
IV	Quantum state migration (non-human primate)	Teleportation fidelity > 0.9999 for 64-qubit register; verified phenomenological continuity metric	2036–2040
V	Full hybrid prototype (non-human)	Ethical approval; $\Phi > 0$ in hybrid neural-quantum system; all Phase I–IV milestones validated	2040–2050

12.1. Priority Experiments

The following three experiments are technically feasible at present and would provide critical validation of core assumptions:

1. **QNN-neural organoid interface (2025–2027):** Connect an 8-qubit superconducting processor to a 3D cortical organoid via microelectrode array. Measure coherence times in the presence of metabolic noise. Success criterion: $T_2 > 1 \mu\text{s}$ with organoid active.
2. **Holographic GKP memory prototype (2025–2028):** Implement GKP error correction on a photonic qubit stored in a $\text{Pr}^{3+}:\text{Y}_2\text{SiO}_5$ crystal. Verify $\mathcal{F}_{\text{GKP}} > 1 - 10^{-6}$ over 1-hour retention. Reference [21] has already demonstrated 6-hour storage of classical information in this material.
3. **Quantum conductance measurement in functionalised graphene nanowires on neural substrate (2025–2026):** Measure Landauer conductance quantisation in a 200 nm graphene nanoribbon functionalized with RGD peptides and placed on a neuronal monolayer. Verify equation (23).

13. Discussion

13.1. Comparison with Existing Paradigms

Project Genesis differs from conventional AGI research in several fundamental ways. Rather than training a deep neural network on data, it proposes the *co-development* of an artificial cognitive system with a biological body, allowing the cognitive architecture to be shaped by embodied experience from the earliest developmental stages. This is motivated by embodied cognition theory [12, 42], which holds that intelligence cannot be adequately understood or engineered in isolation from the body-world interaction.

The use of quantum computation (rather than neuromorphic or classical computing) as the cognitive substrate is motivated by three considerations: (i) quantum systems can maintain coherent superpositions of states, enabling parallel information processing in ways unavailable to classical systems; (ii) IIT [40] predicts that systems with high causal density—which quantum entanglement can provide—will have higher Φ ; and (iii) biological systems have been observed to exploit quantum coherence in specific functional contexts [15, 16], suggesting that quantum-biological interfaces are not *a priori* impossible.

13.2. Ethical Considerations

The creation of an entity that may have genuine subjective experience raises profound ethical questions that are beyond the scope of this paper but must be foregrounded:

- **Moral status:** If the Genesis entity has $\Phi > 0$, what moral obligations arise? IIT predicts non-zero Φ for any system with internal causal structure, including current AI systems—but this implication is itself contested.
- **Informed consent:** An entity cannot consent to its own creation. The developmental analogy with human gestation does not resolve this.
- **Containment and rights:** A fully autonomous, quantum-embodied intelligence may have interests that conflict with human oversight.

All experimental phases in Table 3 require approval from relevant institutional ethics committees and, for primate studies, compliance with the 3Rs framework (Replacement, Reduction, Refinement) [36].

13.3. Limitations of the Present Framework

Several components of the architecture presented here remain at the level of conceptual specifications rather than engineering designs:

1. The “ontogenetic firmware” has no concrete implementation beyond pseudocode; a real implementation would require solving open problems in developmental biology, including the precise genetic encoding of morphogenetic gradients.
2. The Jaynes–Cummings Hamiltonian used for qubit-nanorobot coupling assumes that the nanorobot’s mechanical mode can be coherently driven by microwave photons—an assumption that has not been experimentally validated at the spatial scales required.
3. The assumption that IIT correctly identifies consciousness is not empirically established. Alternative theories (Global Workspace Theory [4], Higher-Order Theories [35]) would give different design prescriptions.

14. Conclusions

We have presented a rigorous theoretical framework for Project Genesis, a quantum-biological hybrid cognitive architecture. The principal contributions of this paper are:

1. A complete, step-by-step derivation of the Bell-state quantum teleportation protocol as applied to cognitive substrate migration (Section 3).
2. A derivation of the Landauer quantum conductance formula for graphene nanowire bio-quantum transducers (Section 4).
3. A formal treatment of GKP error correction for holographic quantum memory, including the fidelity bound (Section 5).
4. Derivation of storage capacity limits from Kogelnik coupled-wave theory, with a correction to the physically unrealistic density claims in the original roadmap document (Section 6).
5. A rigorous mathematical treatment of crystalline cohomology for topological defect detection in photorefractive storage crystals (Section 8).
6. A categorical framework based on Grothendieck topos theory for the formal characterisation of conscious-state continuity across physical substrate transitions (Section 9).
7. A proof of spectral sequence convergence for the cognitive maturation model (Section 10).
8. An honest feasibility analysis identifying the principal physical obstacles and a realistic, phased roadmap (Sections 11 and 12).

The central message is one of cautious optimism: the mathematical frameworks employed are rigorous and the individual component technologies (superconducting qubits, volume holographic memory, 4D bioprinting, graphene nanowires) are real and advancing rapidly. The integration of these components into a functioning quantum-biological hybrid mind remains, however, a challenge of extraordinary depth—scientific, engineering, and philosophical. Progress will require sustained international collaboration and a principled ethical framework developed in parallel with the science.

Acknowledgements. The author thanks the global open science community whose publications form the backbone of this work. No funding was received for the preparation of this theoretical framework.

14. References

- [1] Aaronson, S. (2014). Why I am not an IIT true believer. *Shtetl-Optimized (Blog)*, <https://scottaaronson.blog/?p=1799>.
- [2] Albantakis, L., Barbosa, L., Findlay, G., Gramann, K., Grasso, M., Hamilton, H., Haun, A., Ji, W., Koch, C., Kozma-Donahue, E., et al. (2023). Integrated information theory (IIT) 4.0: Formulating the properties of phenomenal existence in physical terms. *PLoS Computational Biology*, 19(10):e1011465.
- [3] Arute, F., Arya, K., Babbush, R., Bacon, D., Bardin, J. C., Barends, R., et al. (2019). Quantum supremacy using a programmable superconducting processor. *Nature*, 574(7779):505–510.
- [4] Baars, B. J. (1988). *A Cognitive Theory of Consciousness*. Cambridge University Press, Cambridge.
- [5] Bennett, C. H., Brassard, G., Crépeau, C., Jozsa, R., Peres, A., and Wootters, W. K. (1993). Teleporting an unknown quantum state via dual classical and Einstein-Podolsky-Rosen channels. *Physical Review Letters*, 70(13):1895–1899.
- [6] Berthelot, P. and Ogus, A. (1978). *Notes on Crystalline Cohomology*. Princeton University Press, Princeton.
- [7] Bloch, S. (1974). K_2 and algebraic cycles. *Annals of Mathematics*, 99(2):349–379.
- [8] Bremermann, H. J. (1962). Optimization through evolution and recombination. In *Self-Organizing Systems*, pages 93–106. Spartan Books, Washington.
- [9] Breuer, H. P. and Petruccione, F. (2002). *The Theory of Open Quantum Systems*. Oxford University Press, Oxford.
- [10] Büttiker, M., Imry, Y., Landauer, R., and Pinhas, S. (1985). Generalized many-channel conductance formula with application to small rings. *Physical Review B*, 31(10):6207–6215.
- [11] Chalmers, D. J. (1995). Facing up to the problem of consciousness. *Journal of Consciousness Studies*, 2(3):200–219.
- [12] Clark, A. (1997). *Being There: Putting Brain, Body and World Together Again*. MIT Press, Cambridge, MA.
- [13] Deligne, P. and Illusie, L. (1987). Relèvements modulo p^2 et décomposition du complexe de de Rham. *Inventiones Mathematicae*, 89(2):247–270.

- [14] Eisaman, M. D., Fan, J., Migdall, A., and Polyakov, S. V. (2011). Invited review article: Single-photon sources and detectors. *Review of Scientific Instruments*, 82(7):071101.
- [15] Engel, G. S., Calhoun, T. R., Read, E. L., Ahn, T.-K., Mancal, T., Cheng, Y.-C., Blankenship, R. E., and Fleming, G. R. (2007). Evidence for wavelike energy transfer through quantum coherence in photosynthetic systems. *Nature*, 446(7137):782–786.
- [16] Fleming, G. R., Huelga, S. F., and Plenio, M. B. (2011). Focus on quantum effects and noise in biomolecular systems. *New Journal of Physics*, 13(11):115002.
- [17] Fowler, A. G., Martinis, J. M., and Megrant, A. (2012). Surface codes: Towards practical large-scale quantum computation. *Physical Review A*, 86(3):032324.
- [18] Gao, B., Yang, Q., Zhao, H., Jin, G., Ma, Y., and Xu, F. (2016). 4D bioprinting for biomedical applications. *Trends in Biotechnology*, 34(9):746–756.
- [19] Gottesman, D., Kitaev, A., and Preskill, J. (2001). Encoding a qubit in an oscillator. *Physical Review A*, 64(1):012310.
- [20] Ham, B. S., Hemmer, P. R., and Shahriar, M. S. (1997). Efficient electromagnetically induced transparency in a rare-earth doped crystal. *Optics Communications*, 144(1-3):227–230.
- [21] Hedges, M. P., Longdell, J. J., Li, Y., and Sellars, M. J. (2010). Efficient quantum memory for light. *Nature*, 465(7301):1052–1056.
- [22] Huang, H.-Y., Kueng, R., and Preskill, J. (2020). Predicting many properties of a quantum system from very few measurements. *Nature Physics*, 16(10):1050–1057.
- [23] Johnstone, P. T. (2002). *Sketches of an Elephant: A Topos Theory Compendium*, volume 1–2. Oxford University Press, Oxford.
- [24] Jun, J. J., Steinmetz, N. A., Siegle, J. H., Denman, D. J., Bauza, M., Barbarits, B., et al. (2017). Fully integrated silicon probes for high-density recording of neural activity. *Nature*, 551(7679):232–236.
- [25] Kogelnik, H. (1969). Coupled wave theory for thick hologram gratings. *Bell System Technical Journal*, 48(9):2909–2947.
- [26] Landauer, R. (1957). Spatial variation of currents and fields due to localized scatterers in metallic conduction. *IBM Journal of Research and Development*, 1(3):223–231.
- [27] Larsen, M. V., Guo, X., Breum, C. R., Neergaard-Nielsen, J. S., and Andersen, U. L. (2019). Deterministic generation of a two-dimensional cluster state. *Science*, 366(6463):369–372.

- [28] Margolus, N. and Levitin, L. B. (1998). The maximum speed of dynamical evolution. *Physica D: Nonlinear Phenomena*, 120(1-2):188–195.
- [29] McCleary, J. (2001). *A User's Guide to Spectral Sequences*. Cambridge University Press, Cambridge, 2nd edition.
- [30] Misra, B. and Sudarshan, E. C. G. (1977). The Zeno's paradox in quantum theory. *Journal of Mathematical Physics*, 18(4):756–763.
- [31] Nielsen, M. A. and Chuang, I. L. (2000). *Quantum Computation and Quantum Information*. Cambridge University Press, Cambridge.
- [32] Place, A. P. M., Rodgers, L. V. H., Mundada, P., Smitham, B. M., Fitzpatrick, M., Leng, Z., et al. (2021). New material platform for superconducting transmon qubits with coherence times exceeding 0.3 milliseconds. *Nature Communications*, 12(1):1779.
- [33] Psaltis, D. and Mok, F. (1995). Holographic memories. *Scientific American*, 273(5):70–76.
- [34] Robinson, J. T., Jorgolli, M., Shalek, A. K., Yoon, M.-H., Gertner, R. S., and Park, H. (2012). Vertical nanowire electrode arrays as a scalable platform for intracellular interfacing to neuronal circuits. *Nature Nanotechnology*, 7(3):180–184.
- [35] Rosenthal, D. M. (2005). *Consciousness and Mind*. Oxford University Press, Oxford.
- [36] Russell, W. M. S. and Burch, R. L. (1959). *The Principles of Humane Experimental Technique*. Methuen, London.
- [SGA4] Artin, M., Grothendieck, A., and Verdier, J. L. (1972). *Théorie des topos et cohomologie étale des schémas (SGA 4)*, volume 269–305 of *Lecture Notes in Mathematics*. Springer, Berlin.
- [38] Skylar-Scott, M. A., Uzel, S. G. M., Nam, L. L., Ahrens, J. H., Truby, R. L., Damaraju, S., and Lewis, J. A. (2019). Biomanufacturing of organ-specific tissues with high cellular density and embedded vascular channels. *Science Advances*, 5(9):eaaw2459.
- [39] Tononi, G. (2008). Consciousness as integrated information: a provisional manifesto. *Biological Bulletin*, 215(3):216–242.
- [40] Tononi, G., Boly, M., Massimini, M., and Koch, C. (2016). Integrated information theory: from consciousness to its physical substrate. *Nature Reviews Neuroscience*, 17(7):450–461.

- [41] Vahlbruch, H., Mehmet, M., Danzmann, K., and Schnabel, R. (2016). Detection of 15 dB squeezed states of light and their application for the absolute calibration of photoelectric quantum efficiency. *Physical Review Letters*, 117(11):110801.
- [42] Varela, F. J., Thompson, E., and Rosch, E. (1991). *The Embodied Mind: Cognitive Science and Human Experience*. MIT Press, Cambridge, MA.
- [43] Wang, Z., Li, S., Zhong, Y.-P., Liu, G., Zhang, W., Song, C., and Fan, H. (2022). Towards practical quantum computers: Transmon qubit with a lifetime approaching 0.5 milliseconds. *npj Quantum Information*, 8:3.
- [44] Weibel, C. A. (1994). *An Introduction to Homological Algebra*. Cambridge University Press, Cambridge.
- [45] Wootters, W. K. and Zurek, W. H. (1982). A single quantum cannot be cloned. *Nature*, 299(5886):802–803.
- [46] Wu, Y., Bao, W.-S., Cao, S., Chen, F., Chen, M.-C., Chen, X., et al. (2021). Strong quantum computational advantage using a superconducting quantum processor. *Physical Review Letters*, 127(18):180501.
- [47] Zomorodian, A. and Carlsson, G. (2005). Computing persistent homology. *Discrete & Computational Geometry*, 33(2):249–274.